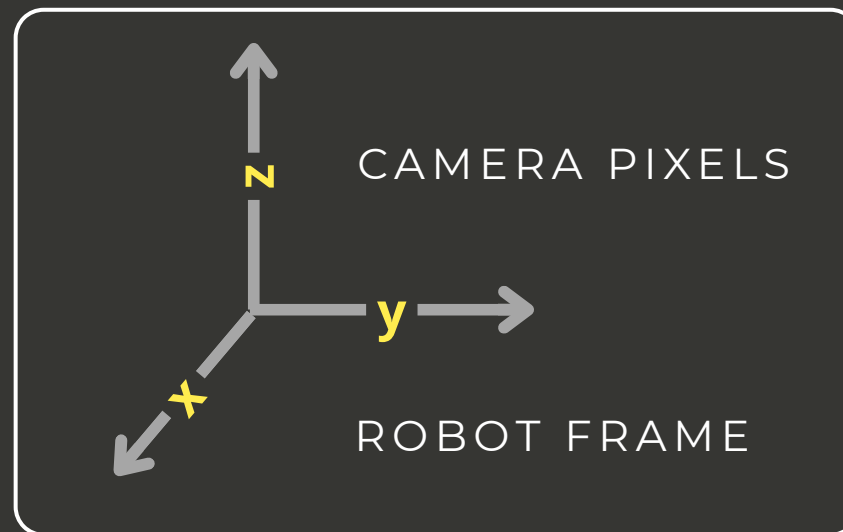


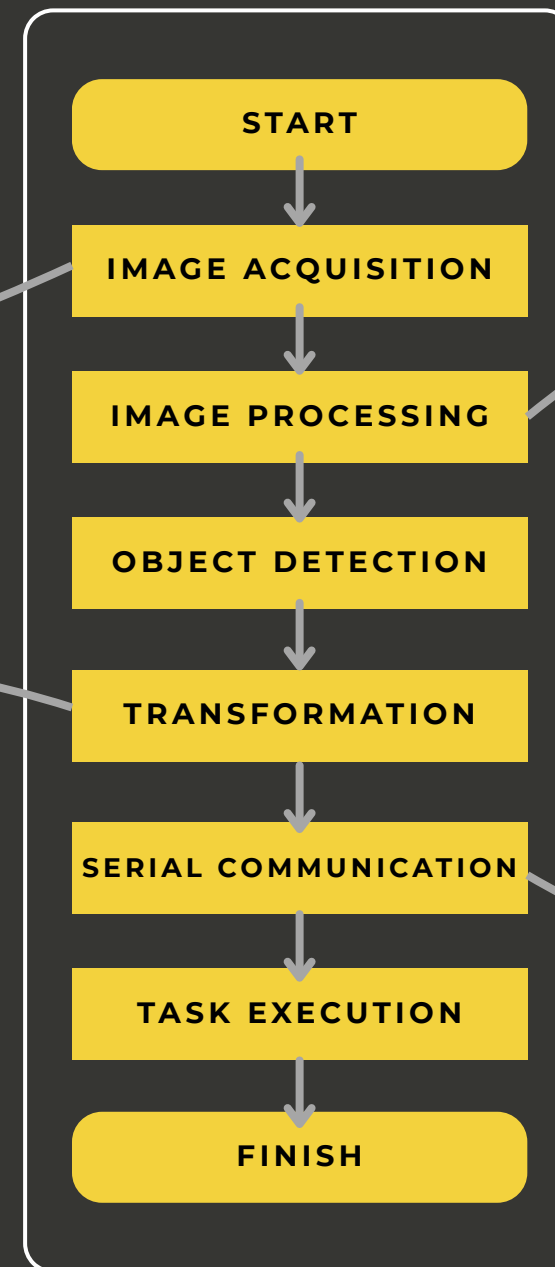
AUTONOMOUS CYLINDER PICK SYSTEM



ESP32-S3
AI CAM



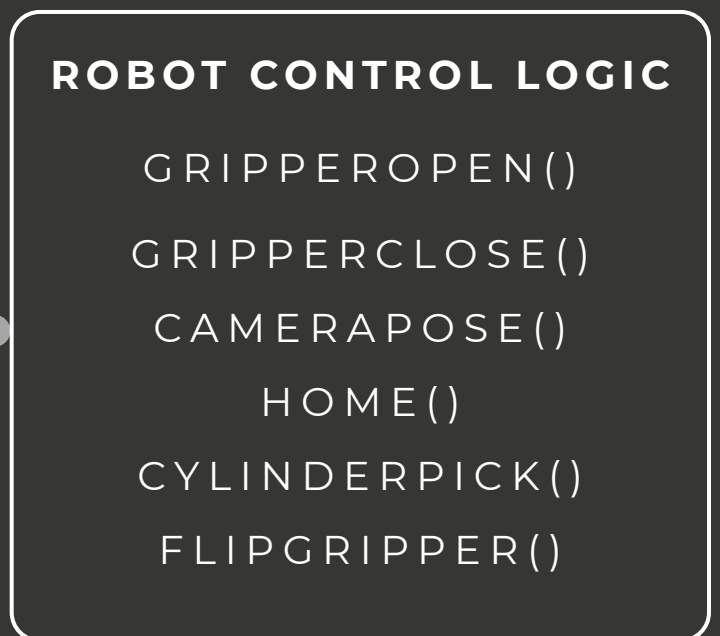
CYLINDER
LOCATION



PROGRAMMING
SCRIPT



SEQUENCING



ROBOT STRUCTURE

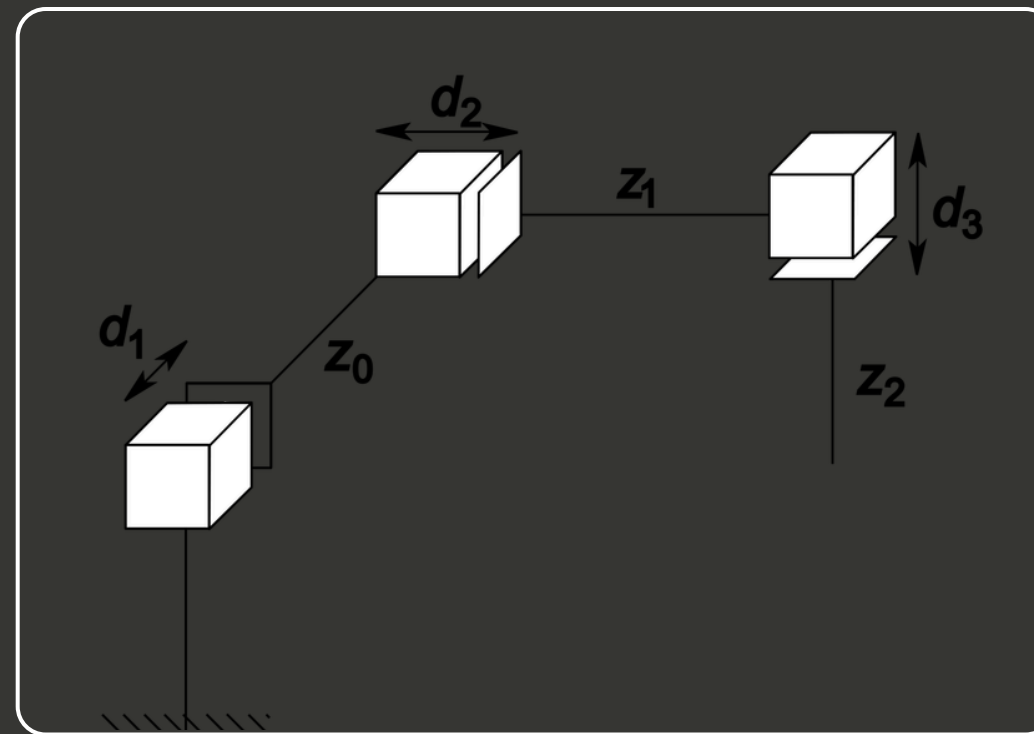
ARCHITECTURE

WHY?
SIMPLE KINEMATICS

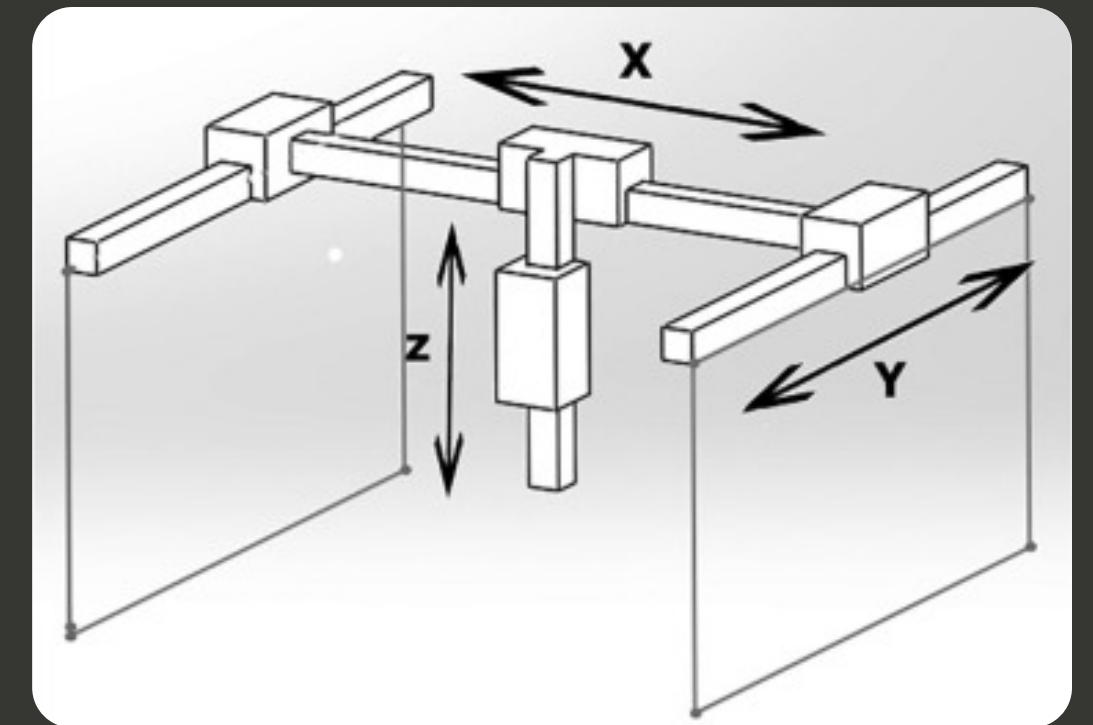
ASSEMBLY

SCREWS AND NUTS
CABLE MANAGEMENT

GANTRY ROBOT



CARTESIAN CONFIGURATION



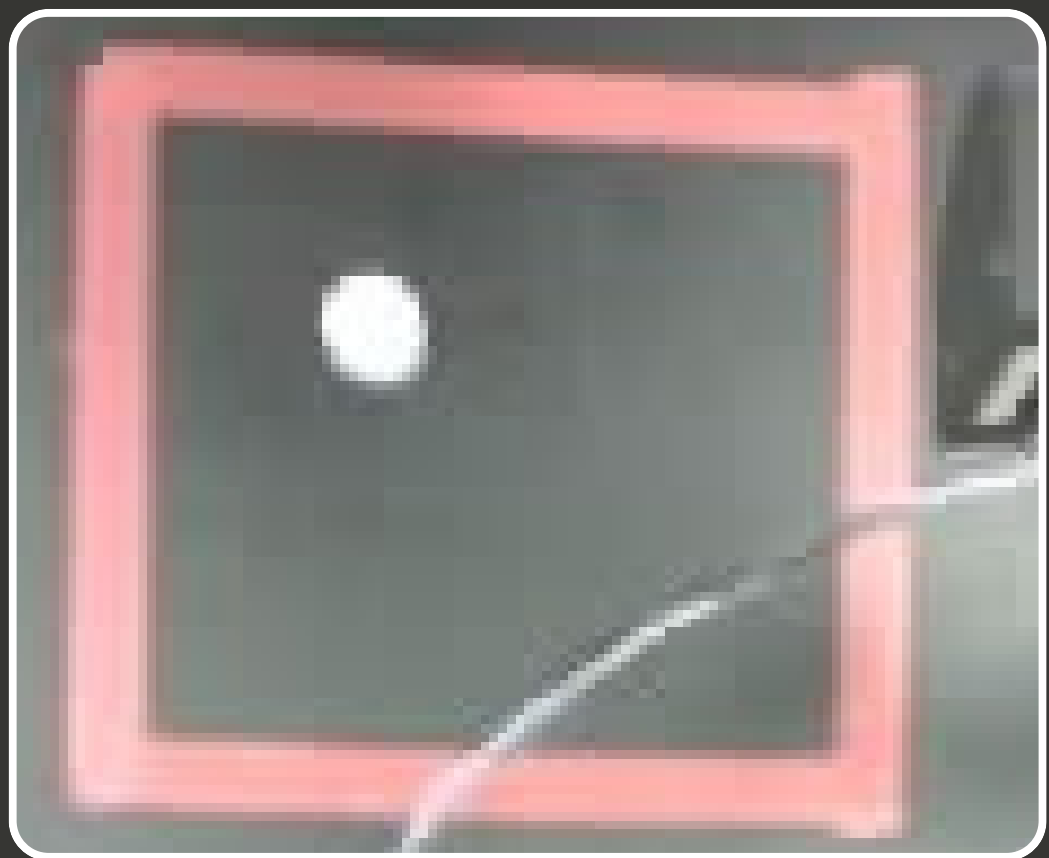
VISION SYSTEM

ESP32-S3 AI CAMERA



WIDE ANGLE
ONLINE STREAM

IMAGE PROCESSING



GAUSSIAN BLUR
HOUGH CIRCLES

OBJECT DETECTION



PIXEL COORDINATES

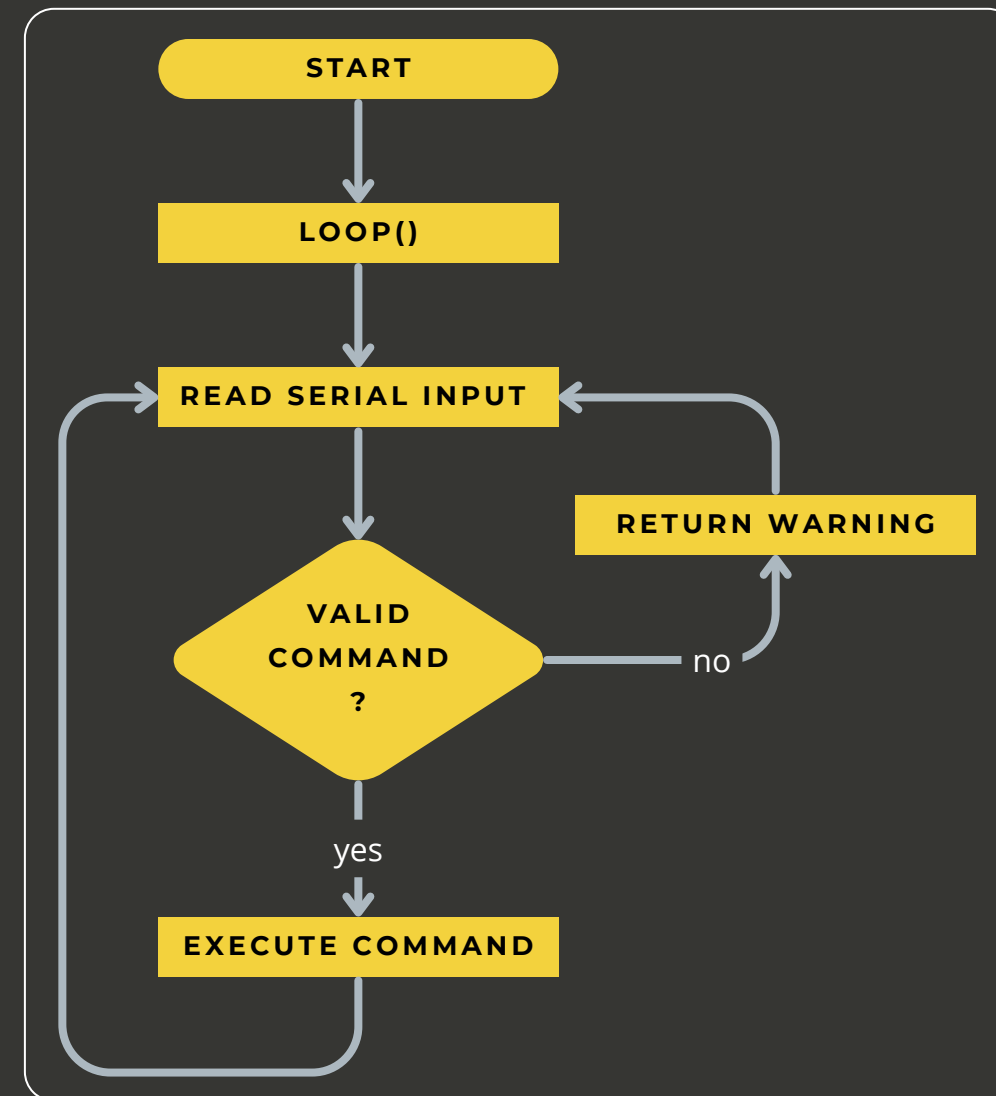
TASK SEQUENCING

SERIAL COMMUNICATION

SHARED WIFI

DEFINED FCUNTIONS
DEBUGGING

ARDUINO LOGIC



PYHTON LOGIC

